

Computer Science - Graduate Level Assessment 1

Instructions

- Answer all questions.

Multiple Choice

Question 1

(2 marks)

_____ is the central node of 802.11 wireless operations.

- A. WPA
- B. Access Point
- C. WAP
- D. Access Port

Question 2

(2 marks)

How does a buffer overflow on the stack facilitate running attacker-injected code?

- A. By overwriting the return address to point to the location of that code
- B. By writing directly to the instruction pointer register the address of the code
- C. By writing directly to %eax the address of the code
- D. By changing the name of the running executable, stored on the stack

Question 3

(2 marks)

Which of the following styles of fuzzer is more likely to explore paths covering every line of code in the following program?

- A. Generational
- B. Blackbox

C. Whitebox

D. Mutation-based

Question 4

(2 marks)

Local caching of files is common in distributed file systems, but it has the disadvantage that

A. Temporary inconsistencies among views of a file by different machines can result

B. The file system is likely to be corrupted when a computer crashes

C. A much higher amount of network traffic results

D. Caching makes file migration impossible

Question 5

(2 marks)

A digital signature needs a

A. Private-key system

B. Shared-key system

C. Public-key system

D. All of them

True / False

Question 1

(2 marks)

Encryption and decryption ensure both confidentiality and integrity of the data being transmitted.

True

False

Question 2

(2 marks)

The addition of the 8-bit two's-complement numbers 10000001 and 10101010 results in overflow

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True

False

Question 3

(2 marks)

A virus is a type of malware that operates solely as a standalone infected program without any hidden components.

True

False

Question 4

(2 marks)

All bit strings with an even number of zeros can be described with a regular expression.

True

False

Question 5

(2 marks)

The solution to the recurrence $f(2N + 1) = f(2N) = f(N) + \log N$ is $O((\log N)^2)$.

True

False

Long Form

Question 1

(10 marks)

Write a python function to find the nth digit in the proper fraction of two given numbers.

Type your answer here...



Question 2

(10 marks)

Write a function to group the 1 st elements on the basis of 2 nd elements in the given tuple list.

Type your answer here...



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